

Dao Xuan Tan

MLOps Engineer

📍 Ho Chi Minh, Vietnam

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Summary

MLOps engineer building and optimizing production ML systems across model lifecycle, data and embedding pipelines, low-latency model serving, evaluation metrics, and monitoring. Track record of profiling large-scale pipelines, hunting bottlenecks, and cutting latency at scale on strong Python and algorithms and data-structures foundation; quick to pick up new languages and technology.

Experience

Pi Associates, Software Engineer, Compute Platform

Python, asyncio, multiprocessing, Kafka, PostgreSQL, Redis, Docker, Kubernetes, GCP, Grafana

Ho Chi Minh

Jan 2026 – present

7 months

- Own internal **Python** compute platform running **tens of thousands** of quant model backtest and evaluation **jobs daily** for about **20** researchers
- Investigated pipeline, **identified bottlenecks**, and cut runtime from **minutes to under 10 seconds** on **1 to 5 GB** per run through **parallelism**, vectorization, and algorithmic improvements
- Designed **concurrent execution model** (**asyncio** plus **process pool**) to maximize **job throughput** under heavy parallel load
- Built **Kafka** producer streaming model signals to downstream consumers, production **data-pipeline component** feeding real-time systems
- Ran jobs as Dockerized **Kubernetes** workloads across **GCP** and on-premise, with **GitHub Actions** CI/CD and **Grafana** monitoring
- Hardened reliability with stronger error handling and input validation, reducing failed runs and manual debugging

AcamarAI, AI/ML Engineer

Python, vLLM, model serving, data pipelines, evaluation, LoRA, FastAPI, PyTorch

Ho Chi Minh

June 2025 – Jan 2026

8 months

- Served self-hosted **Qwen** models with **vLLM** on single **A100 80GB** behind **OpenAI-compatible API**, delivering **low-latency serving** (about **1s TTFT**, **200 tok/s** aggregate)
- Built ingestion and **embedding data pipelines** (chunking strategies, **BGE** embeddings, **ColPali** for document images) feeding self-hosted vector store over **millions of vectors**
- Developed **evaluation metrics and harnesses**, **recall@k** and **LLM-as-judge**, to measure and improve model output quality
- Fine-tuned task-specific models with **LoRA** and shipped **3 of 14** core AI features for Fortune 500 beta users via **async FastAPI** with **SSE** streaming

Skills

Model Serving & Optimization: vLLM, self-hosted Qwen, OpenAI-compatible APIs, low-latency and high-throughput serving, LoRA fine-tuning, GPU (A100)

Data & ML Pipelines: ingestion and embedding pipelines, Kafka streaming, batch and compute jobs, vector indexing (Qdrant), evaluation harnesses (recall@k, LLM-as-judge)

Performance & Scale: multiprocessing and parallelism (asyncio + process pool), vectorization, profiling, latency and throughput optimization, algorithms and data structures

Backend & APIs: Python, FastAPI, asyncio, SSE streaming, REST APIs, Pydantic validation, PostgreSQL, Redis

Infra & MLOps: Docker, Kubernetes, GCP, on-premise, GitHub Actions CI/CD, Grafana

Tools: AI-assisted development (Claude Code, Cursor, Codex), Git, uv

Languages: Vietnamese (Native), English (IELTS 6.5)

Education

Ho Chi Minh University of Science, Mathematics & Computer Science (Data Science)

Ho Chi Minh City

- GPA: 8.74/10.0 (3.74/4.0)

Oct 2022 – Sept 2026